

1.A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system. (20 Marks)

Code:

```
ping -c 4 192.168.1.10
```

```
nmap 192.168.1.10
```

```
nmap -p- 192.168.1.10
```

```
sudo nmap -sS 192.168.1.10
```

```
sudo nmap -sV 192.168.1.10
```

```
sudo nmap -O 192.168.1.10
```

```
sudo nmap -sC 192.168.1.10
```

```
sudo nmap -A 192.168.1.10
```

```
sudo nmap -Pn 192.168.1.10
```

```
sudo nmap -A -oN nmap_report.txt 192.168.1.10
```

```
cat nmap_report.txt
```

Explanation:

ping -c 4 192.168.1.10

Checks whether the target system is reachable by sending four ICMP echo requests.

nmap 192.168.1.10

Performs a basic scan to identify open ports on the target system.

nmap -p- 192.168.1.10

Scans all TCP ports (1–65535) on the target system.

sudo nmap -sS 192.168.1.10

Performs a TCP SYN (Stealth) scan to detect open ports.

sudo nmap -sV 192.168.1.10

Detects the versions of services running on the open ports.

sudo nmap -O 192.168.1.10

Detects the operating system of the target machine.

sudo nmap -sC 192.168.1.10

Runs the default Nmap Scripting Engine (NSE) scripts to gather additional information.

sudo nmap -A 192.168.1.10

Performs an aggressive scan including OS detection, service version detection, script scanning, and traceroute.

sudo nmap -Pn 192.168.1.10

Skips host discovery and assumes the target host is online.

sudo nmap -A -oN nmap_report.txt 192.168.1.10

Performs an aggressive scan and saves the scan results in a file named nmap_report.txt.

cat nmap_report.txt

Displays the contents of the saved Nmap scan report.

2.A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.(20 Mark).

Code:

```
sudo systemctl start gophish
```

```
cd /opt/gophish
```

```
./gophish
```

firefox http://127.0.0.1:3333

firefox http://127.0.0.1:8080

sudo netstat -tulnp | grep gophish

Explanation:

sudo systemctl start gophish

Starts the GoPhish phishing simulation service.

cd /opt/gophish

Changes the current directory to the GoPhish installation folder.

./gophish

Launches the GoPhish application.

firefox http://127.0.0.1:3333

Opens the GoPhish administrator dashboard to create phishing campaigns, email templates, landing pages, and view reports.

firefox http://127.0.0.1:8080

Opens the phishing landing page used for user awareness testing.

sudo netstat -tulnp | grep gophish

Verifies that the GoPhish service is running and displays the listening ports.

3.A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure..(10 Marks)

Code:

dig @192.168.1.10 example.com

dig @192.168.1.10 example.com ANY

dig @192.168.1.10 example.com MX

dig @192.168.1.10 example.com NS

dig @192.168.1.10 example.com AXFR

Explanation:

dig @192.168.1.10 example.com

Queries the DNS server for the default DNS records of the domain.

dig @192.168.1.10 example.com ANY

Requests all available DNS records for the domain.

dig @192.168.1.10 example.com MX

Displays the Mail Exchange (MX) records used for email routing.

dig @192.168.1.10 example.com NS

Displays the authoritative Name Server (NS) records for the domain.

dig @192.168.1.10 example.com AXFR

Attempts a DNS Zone Transfer (AXFR). If successful, it reveals all DNS records, including internal hostnames, mail servers, subdomains, and other sensitive DNS information, indicating a DNS server misconfiguration.